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ORIGINAL ARTICLE

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Thermal bioclimatic conditions and patterns of behaviour in an urban park in Göteborg, Sweden

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Abstract People in urban areas frequently use parks for recreation and outdoor activities. Owing to the complexity of the outdoor environment, there have only been a few attempts to understand the effect of the thermal environment on people's use of outdoor spaces. This paper therefore seeks to determine the relationship between the thermal environment, park use and behavioural patterns in an urban area of Sweden. The methods used include structured interviews, unobtrusive observations of the naturally occurring behaviour and simultaneous measurements of thermal comfort variables, i.e., air temperature, air humidity, wind speed and global radiation. The thermal environment is investigated through the mean radiant temperature (T_{mrt}) and the predicted mean vote (PMV) index. The outcome is compared to the subjective behaviour and thermal sensation of the interviewees. It is found that the thermal environment, access and design are important factors in the use of the park. In order to continue to use the park when the thermal conditions become too cold or too hot for comfort, people improve their comfort conditions by modifying their clothing and by choosing the most supportive thermal opportunities available within the place. The study also shows that psychological aspects such as time of exposure, expectations, experience and perceived control may influence the subjective assessment. Comparison between the thermal sensation of the interviewees and the thermal sensation assessed by the PMV index indicates that steady-state models such as the PMV index may not be appropriate for the assessment of short-term outdoor thermal comfort, mainly because they are unable to analyse transient exposure.

Keywords Thermal comfort · Human behaviour · Structured interviews · Observations · Urban park

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Introduction

Urban parks play an important role in people's recreational life and outdoor activities. A comfortable climate is an important prerequisite for outdoor relaxation. The temporal and spatial complexity of the outdoor environment has led to only a few attempts being made to understand the effect of thermal comfort conditions on people's use of outdoor spaces.

Since the outdoor thermal environment may not be comfortable all the time, the various microclimates created offer individuals control in overcoming thermal discomfort. People tend to adapt to the ambient thermal conditions by modifying clothing and activity pattern in order to continue their activities and routines (e.g. Nasar and Yurdakul 1990; Gehl 1996; Donaldson et al. 2001; Nikolopoulou et al. 2001; Parsons 2002). This may be done consciously or unconsciously.

Besides physical adaptation, psychological aspects such as expectations, available choice, environmental stimulation, thermal history and memory are important factors in the subjective assessment of outdoor thermal comfort (Nikolopoulou et al. 2001; Nikolopoulou and Steemers 2003). In particular, the expectation of a specific thermal condition is a major aspect of subjective assessment and satisfaction (Höppe and Seidl 1991; Höppe 2002). Another important aspect that must be considered when investigating the outdoor climate is the time of exposure to the outdoor environment (Höppe 2002).

People in industrialised countries spend on average less than 10% of their time outdoors (Leech et al. 2000). This relatively short time means that thermal steady state is hardly ever reached (Höppe 2002). According to simulations of temporal courses of thermophysiological parameters characterising thermal comfort and potential health risks, performed by Höppe (2002), outdoor thermal indices, based on steady-state energy-balance models of the human body, tend to overestimate discomfort. The simulations also show that the discrepancy is larger for cool outdoor conditions than for warm conditions. This is due to the fact that the body temperature adapts to heat, by vasodilatation, much faster than it adapts to cold, by vasoconstriction. The author uses these findings to emphasise that steady-state indices, such as the predicted mean vote (PMV) index (Fanger 1970) and the physiological equivalent temperature (Höppe 1999) may not be appropriate for the assessment of outdoor thermal comfort, especially that associated with short-term exposure to cold conditions.

The climate of urban parks has been exhaustively investigated in Göteborg during the last decades; emphasis has been placed on air temperature, air humidity and wind pattern (Lindqvist 1992; Eliasson 1996; Upmanis et al. 1998; Upmanis and Chen 1999; Eliasson and Upmanis 2000). The aim of the present study is to determine the relationship between thermal environment and park use in an urban area of Sweden. Patterns of behaviour related to sun and shade and the corresponding level of thermal comfort are discussed. In addition, the thermal sensation assessed by the PMV index (Fanger 1970), is compared to the thermal sensation experienced by the interviewees and problems are identified. Both structured interviews and unobtrusive observations of the naturally occurring behaviour are performed while the thermal comfort variables are simultaneously measured. The mean radiant temperature (T_{mrt}) and PMV are calculated with the aid of the PC program RayMan (Matzarakis et al. 2000; Matzarakis 2002).

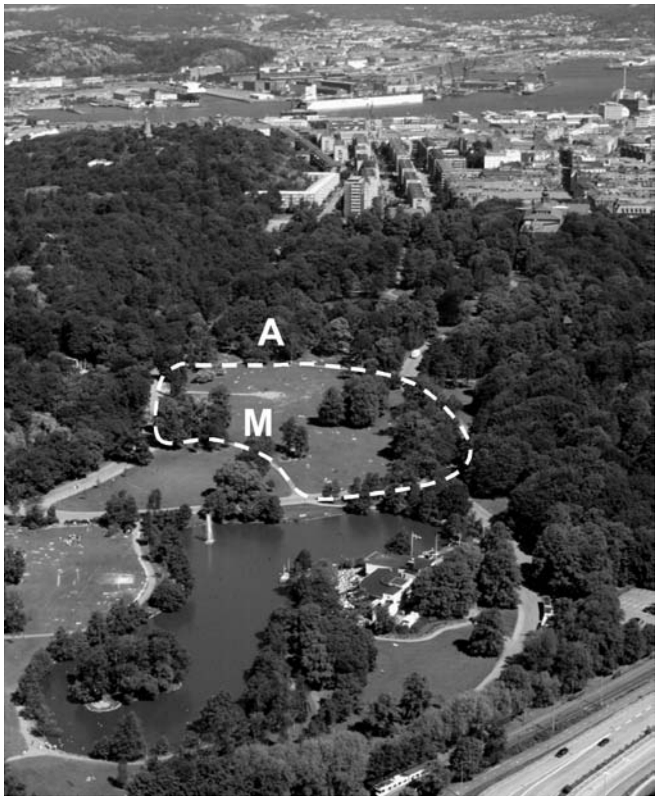


Fig. 1 Photo showing Slottsskogen, the largest park (156 ha) in Göteborg. The study area is marked with a *dashed white line*. Photographs were taken at *A* and the meteorological station is located at *M*

Materials and methods

Description of the area

The metropolitan area of Göteborg (57°42′N, 11°58′E) is located on the west coast of Sweden. It is the second largest city in Sweden with a population of 500,000. Several natural green spaces and parks are located in the city centre, with Slottsskogen (156 ha) being the largest urban park (Fig. 1). The park is a mixture of different land types, such as wooded hills, grass surfaces with scattered trees and bushes, houses, dams, roads and paved footpaths. The park, which is surrounded by built-up areas, is very popular; it is visited by the public all year around, but most frequently in summer.

A 2.5-ha area of grass, with scattered trees and greenery, offering both sun and shade inside the park was selected for this study (Fig. 1). A distinction must be made between route and resting-places when outdoor comfort conditions are being investigated (Nikolopoulou et al. 2001). In contrast to route-places, where discomfort will not cause serious distress since the time of exposure to the specific environmental conditions is short, discomfort in resting-places may distress people, thus leading them to avoid using these areas. The area selected for the study is a typical resting-place which people visit to relax, sunbathe, have a picnic, meet friends and play outdoor games.

Field surveys

Field surveys were performed between July and October 2002. The surveys took place on weekdays between 1 p.m. and 3 p.m. It is at this time of the day that both air temperature and solar radiation reach their approximate daily maximum and the park has the most visitors. Both structured interviews and unobtrusive observations of the naturally occurring behaviour were conducted. At the same time, the air temperature, relative humidity, global radiation and wind speed were measured. Field surveys were only performed on days without precipitation. The days on which precipitation occurred were excluded from the analysis.

Information about people's thermal sensation (actual sensation vote, ASV), general state of health, thermal history (i.e., time spent in the area), estimated air temperature, housing area, reason for their visit and opinions on the design of the park were obtained by using questionnaires. The interviews took place in the sun. Interviewees were asked to report their ASV on the basis of a seven-step psychophysical scale, i.e. hot, warm, slightly warm, neutral, slightly cool, cool and cold (Fanger 1970). Their general state of health was reported, using another seven-step scale where a value of zero represents "as usual", positive values "better than usual" and negative values "poorer than usual" (A. Matzarakis, Meteorological Institute, University of Freiburg, Germany, personal communication). Additional information about demographic factors (such as gender, age, length and weight), clothing (e.g. shorts and T-shirt, bikini, long-sleeved sweater and trousers etc.) and activity (standing, sitting and lying) was noted. The clothing was later denoted by a clothing insulation value (clo) according to Fanger (1970), while the estimations of heat produced by activity (metabolic heat) were made according to Guideline 3787 of the German Engineering Society (VDI 1998). Accurate estimations of heat produced by activity and clothing insulation are necessary if a precise assessment of comfort is to be obtained, since the PMV index is particularly sensitive to these parameters (Olesen and Parsons 2002; Havenith et al. 2002). A total of 300 people were asked, of whom 95% were willing to be interviewed. The subjects were selected according to a scheme, in order to obtain as even a distribution as possible of gender, age and activity.

Table 1 Type of measurement, parameter, instrument and measuring height

Type of measurement	Parameters measured	Instrument	Accuracy	Measuring height (m)
Meteorological station	Air temperature	Rotronic YA-100	<±0.2 °C	4
	Relative humidity	Rotronic YA-100	<±1 %	4
	Wind speed	Vaisala WAA 15	±0.1 m s^{-1}	5
	Global radiation	Skye SP1110	±5 %	5
Mobile station	Air temperature	Almemo 2290-8	±0.1 °C	1.1
	Relative humidity	Almemo 2290-8	±2 %	1.1

Table 2 Minima, maxima and means for the measured meteorological parameters: wind speed, air temperature, relative humidity and global radiation

Meteorological parameter	Minimum	Maximum	Mean
Wind speed at 5 m (m s^{-2})	0.4	3.5	1.6
Wind speed at 1.1 m (m s^{-1})	0.3	2.3	1.1
Air temperature (°C)	6.4	29.0	21.9
Relative humidity (%)	36.7	87.8	58.7
Global radiation (Wm^{-2})	73	882	556

Table 3 Predicted mean vote (*PMV*), thermal sensation and stress levels (VDI 1998)

PMV	Thermal sensation	Stress level	Physiological effect
	Cold	Strong	Cold stress
–2.5	Cool	Moderate	Cold stress
–1.5	Slightly cool	Slight	Cold stress
–0.5	Comfortable	No stress	Cold stress
0.5	Slightly warm	Slight	Heat stress
1.5	Warm	Strong	Heat stress
2.5	Hot	Great	Heat stress

Interviews are an effective method of uncovering thoughts, preferences and intentions, but they can be both reactive and unrealistic (Nasar and Yurdakul 1990). Direct and unobtrusive observations of naturally occurring behaviour may therefore be preferable to interviews (Baker 1968). Thus a total of 50 unobtrusive observations of the total attendance and in situ activities were performed in connection with the interviews. The in situ activities were classified into the same three categories as above. The number of visitors in the sunny and shaded areas within each of these three categories was also noted. Since the number of visitors during weekdays is highly dependent on holidays, one summer month was chosen for this purpose, i.e. July. Swedish schools are closed in July and most people are on holiday at this time. The number of individuals visiting the designated place, as well as those present in the sunny and shaded areas were counted at 1 p.m., 2 p.m. and 3p.m. Only people using the study area as a resting-place were included in the study, i.e. people who were simply passing through the area were excluded. Photographs were taken at point A (Fig. 1) for further documentation of each observation. The photographic technique has been applied in many studies to record the naturally occurring behaviour in different urban environments (e.g. Nasar and Yurdakul 1990; Gehl 1968).

A meteorological station (M), located within the study area (Fig. 1) and equipped according to Table 1, was used to measure the thermal comfort variables. The heights were chosen for practical reasons. As the measuring height should always be 1.1 m above the ground, which corresponds to the average height of the centre of gravity for adults (Mayer and Höppe 1987), air temperature and relative humidity were also measured at this level during the field surveys, using a mobile station. Wind speed at 5 m was recalculated at 1.1 m by using Sverdrup's power law with an exponent of 0.28 (Davenport 1965). The minima, maxima and means of the measured values are given in Table 2.

Predicted mean vote index

The thermal comfort index used was the predicted mean vote (PMV). This index, calculated from Fanger's (1970) comfort equation, is one of the most frequently used indices in human biometeorology. Through parameterisation of the complex radiation fluxes by Jendritzky et al. (1979), Fanger's model has become applicable to outdoor situations under the name "Klima-Michel model" (KMM). Six input parameters are required for comfort assessment: four climatic parameters and two parameters related to the occupants of the environment. The climatic parameters are air temperature, vapour pressure, mean radiant temperature and air speed, while the environmental occupation parameters are metabolic heat production and clothing insulation. The PMV index predicts the expected comfort vote on a seven-step psychophysical scale (Fanger 1970) according to Table 3.

RayMan

The mean radiant temperature (T_{mrt}) and PMV were calculated, using the PC modelling programme RayMan (Matzarakis et al. 2000; Matzarakis 2002). T_{mrt}, which is required in the human energy balance model and for comfort assessment, is defined as the uniform temperature of a surrounding surface emitting black-body radiation (emission coefficient $\varepsilon = 1$), which results in the same radiation energy gain of a human body as the prevailing radiation fluxes (Matzarakis et al. 2000). T_{mrt} is the most important meteorological input parameter for human energy balance during summer weather conditions and thus has a strong influence on thermophysiological indices such as the PMV (Mayer 1993). The RayMan model, developed according to Guideline 3787 of the German Engineering Society (VDI 1998), calculates the radiation flux within urban structures on the basis of parameters that include air temperature, air humidity, degree of cloud cover, air transparency, time of day and year, albedo of the surrounding surfaces and their solid-angle proportions (VDI 1994). However, measurements of T_{mrt} can be directly used as input to the model when available. In this paper global radiation measurements were used as input to the model instead of the degree of cloud cover. The T_{mrt} and PMV were compared to the subjective behaviour and thermal sensation of the interviewees.

Results and discussion

Park use

People of all ages, but mainly between the ages of 20 and 40 years, visited the study area for a few hours. According to the interviews, the majority of the subjects completing the questionnaire feel better or much better than usual when visiting the park (Fig. 2). Furthermore, opinions on

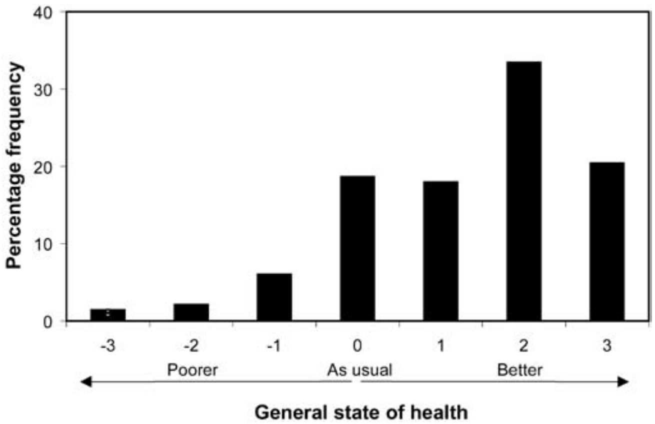


Fig. 2 Percentage frequency distribution for the interviewees' general state of health, where a value of 0 represents "as usual", positive values "better than usual" and negative values "poorer than usual"

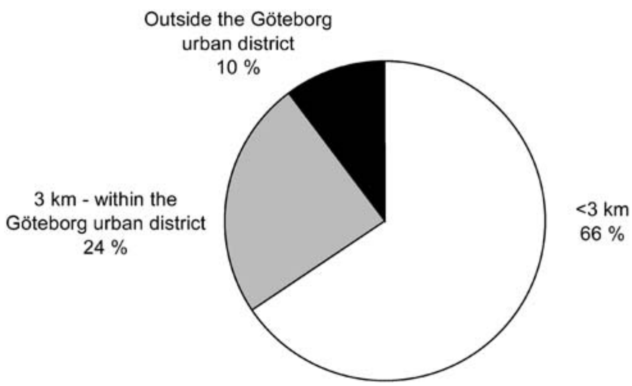


Fig. 3 Accessibility to the study area measured by the distance from the park to the interviewee's residence

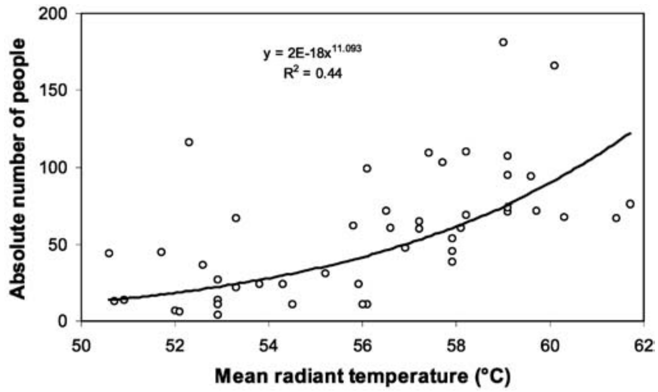


Fig. 4 Absolute number of people present in the study area along with mean radiant temperature (T_{mrt})

Fig. 5 **a** Photographs showing the park in use during a summer day in July, with an ambient air temperature of 28.3 °C, a global radiation of 665 Wm^{-2} and a mean radiant temperature of 59 °C. The number of visitors at the time was 181. **b** Photograph showing the park in use during a summer day in July, with an ambient air temperature of 17.9 °C, global radiation of 414 Wm^{-2}, mean radiant temperature of 51 °C. The number of visitors at the time was 14

the design of the park are thoroughly positive. When people were asked in more detail to say what they liked about the park they stated its size, the combination of wooded and large open areas, the peacefulness and its suitability for many different activities. However, accessibility to the park was stated as one of the most important factors. This is further illustrated in Fig. 3, which shows that the majority (66%) of the visitors live within a short distance from the park (i.e., <3 km), even though people from all sections of Göteborg's urban district as well as outside the city are represented among the visitors.

In order to examine whether different thermal comfort conditions influence the use of the park the number of individuals making use of the designated area were counted. As expected, warm and sunny conditions are shown to be important factors in the use of the park. The absolute number of people present in the study area is shown in Fig. 4 as a function of the mean radiant temperature (T_{mrt}). Even though large variations exist, an increase in the number of people is observed with increasing T_{mrt}. Examples of use of the park during different weather conditions are shown in Fig. 5.

Physical adaptation

In order to continue to use the park when the thermal environment becomes too cold or too hot for comfort, people improve their level of comfort, either consciously or unconsciously, by modifying their clothing (Fig. 6). However, the level of activity, modesty, acceptance and clothing design will determine the amount of clothing worn (Parsons 2002). Another way to avoid discomfort is

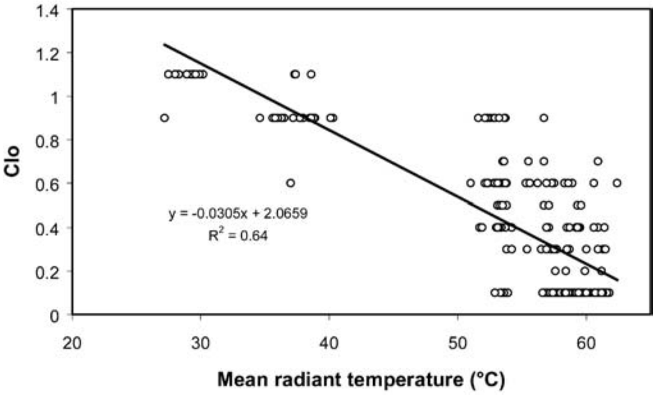


Fig. 6 Variation of clothing insulation (clo) with mean radiant temperature (T_{mrt})

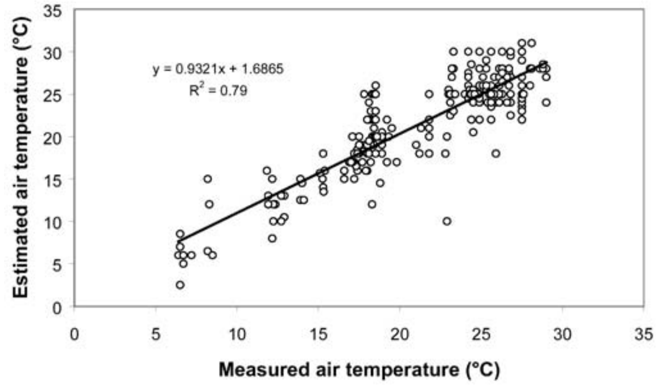


Fig. 8 Air temperature estimated by interviewees (T_{e}) plotted against measured air temperature (T_{a})

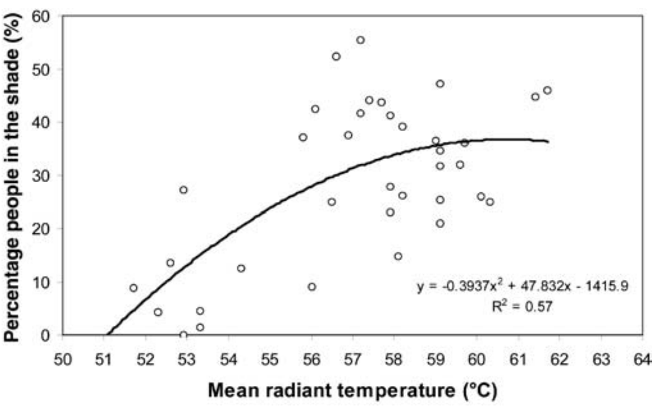


Fig. 7 Distribution of the percentage of people in the shade with mean radiant temperature (T_{mrt})

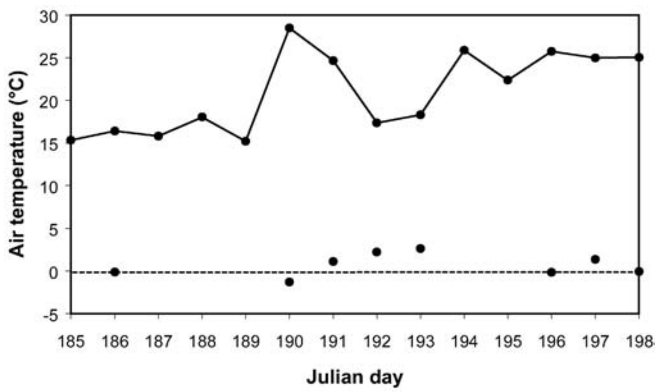


Fig. 9 Air temperature at 2 p.m. (*solid line*) and mean difference between estimated and ambient air temperature ($\Delta T_{\mathrm{e-a}}$) between 4 and 17 July 2002 (Julian days 185–198). The number of interviews for the different days varied between 10 and 20

to change patterns of behaviour by choosing the most supportive thermal environment opportunities available within the area. The results presented in Fig. 7 show that most people visit the park to be out in the sun, but when the weather gets too hot some people move out of direct sunlight and into the shade. The variety of microclimates within the area, i.e., sun and shade, as well as sheltered and exposed spaces enables great individual control in overcoming thermal discomfort, since such spaces make it easier for people to change between different thermal conditions in order to achieve or maintain thermal comfort. The fact that 25%–55% of the visitors (i.e., 15–66 people) seek the shade during warm and sunny conditions demonstrates the importance of creating both sunny and shaded sub-spaces in outdoor public places.

The results confirm that people adapt in order to improve the relation between the thermal environment and their particular requirements: altering the clothing level, posture and position etc (e.g. Gehl 1968, 1996; Nikolopoulou et al. 2001; Zacharias et al. 2001).

Thermal sensation

The Swedish weather and season are often limiting factors for outdoor activities. People are very aware of the weather, so much so that it is a common topic of conversation. Interviewees were asked to estimate the ambient air temperature (T_{a}). As shown in Fig. 8 people tend to overestimate the air temperature slightly. A closer examination of the pattern reveals that during long periods with similar weather conditions the mean difference between the estimated and ambient air temperatures ($\Delta T_{\mathrm{e-a}}$) is close to zero (Fig. 9). However, people tend to underestimate the air temperature when the weather changes and becomes warmer. The opposite occurs when the weather becomes cooler. This indicates that short-term experiences may affect people's thermal sensitivity. The role of experience in satisfaction with and thermal sensitivity to the outdoor environment has also been discussed by Nikolopoulou et al. (2001), Nagara et al. (1996) and Nikolopoulou and Steemers (2003), among others.

The interviewees' thermal sensation (ASV) is compared to that assessed by the PMV, taking all the thermal

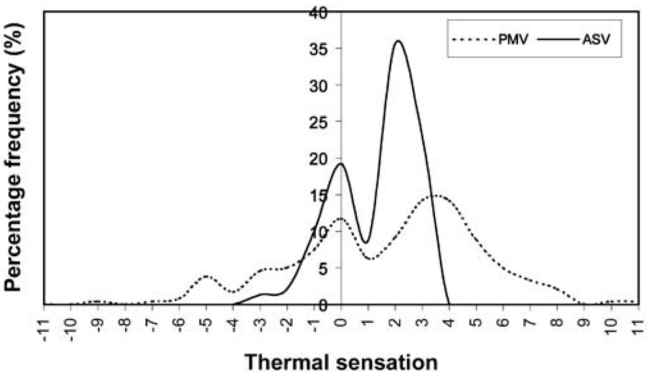


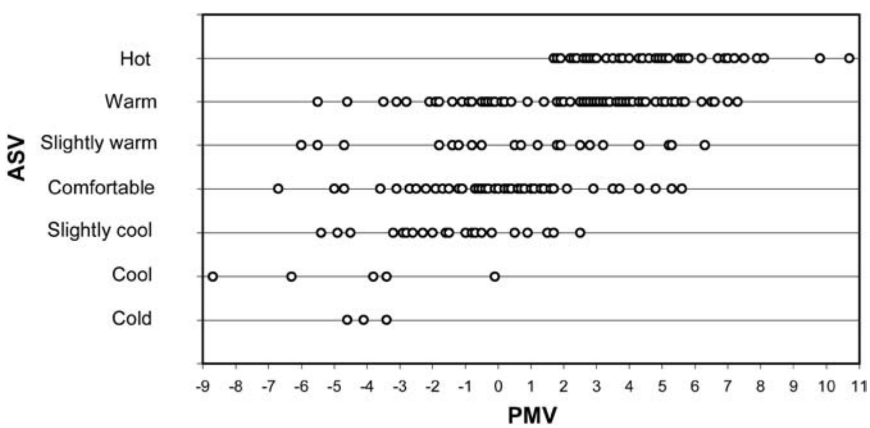
Fig. 10 Percentage frequency distribution for thermal sensation assessed by the predicted mean vote (*PMV*) index and the thermal sensation of the interviewees (*ASV*)

comfort variables recorded during the interview, as well as the demographic factors, clothing and activity for each interviewee into account. It is apparent that there is a distinct discrepancy in percentage frequency distribution between the two curves (Fig. 10). The majority (59%) of the interviewees voted for warm (2) or hot (3) thermal conditions, 38% for acceptable comfort conditions, i.e., slightly cool (–1), comfortable (0) and slightly warm (1), whereas only 3% voted for cool (–2) or cold (–3) thermal conditions. The corresponding PMV curve is also skewed towards the warm zone. According to the PMV index only 26% of the interviewees were within the acceptable theoretical comfort conditions, 23% within warm (2) or hot (3) and 10% within cool (–2) or cold (–3) thermal conditions. It is worth noting that 41% of the interviewees were outside the range of Fanger's model (<–3 and >3). Although opinions on the weather are individual, as illustrated in Fig. 11, these results suggest a number of things. The most apparent is that people voluntarily expose themselves to heat that is well outside theoretically comfortable conditions when visiting the park; this implies that psychological aspects in terms of expectancy may be of significance to the subjective assessment and satisfaction. Another example of an outdoor place where people voluntarily expose themselves to very adverse thermal conditions is the beach (Höppe and Seidl 1991; de Freitas 1985).

To examine whether the PMV index provides a realistic assessment of cold, warm, short and long-term outdoor exposure, the difference between the ASV and the thermal sensation assessed by the PMV index was analysed according to time spent in the park and ambient temperature (T_a). According to the interviews most people reported being comfortable between 18 °C and 19 °C. Thus, the material was divided into $T_a \leq 18.5$ °C and $T_a \geq 18.5$ °C. The PMV values in the ranges <–3 and >+3 were excluded from the analysis since they are outside the range of Fanger's model. The result presented in Fig. 12 shows that there are difficulties associated with using the PMV index to provide realistic assessment data, particularly for short-term outdoor exposure. However, no statistically significant difference between long- and short-term exposure could be assumed at the 0.05 significance level because of the limited amount of data. A plausible explanation for the discrepancy may be that the PMV model is based on indoor data and that the thermal preference value may vary between outdoors and indoors because of the wide range of complex factors influencing the human response outdoors (e.g. Brager and de Dear 1998). Furthermore, Fanger's model is intended to represent what a hypothetical "average" person will feel (Fanger 1970).

Individual differences between people (inter-individual variance), as well as day-to-day differences (intra-individual variance) both frequently vary by the order of one scale value on a seven-point thermal sensation scale (Fountain et al. 1996). However, the fact that the discrepancy tends to decrease with time spent in the park could be explained by the fact that it takes time for the body to adapt when the thermal conditions are changed. Since steady-state state models such as the PMV are confined to situations in which people stay outdoors for a long-time, the thermal sensation of the interviewees may differ form the thermal sensation assessed by the index when the outdoor exposure is short (Höppe 2002).

Fig. 11 Thermal sensation of the interviewees (*ASV*) plotted against thermal sensation assessed by the predicted mean vote (*PMV*) index



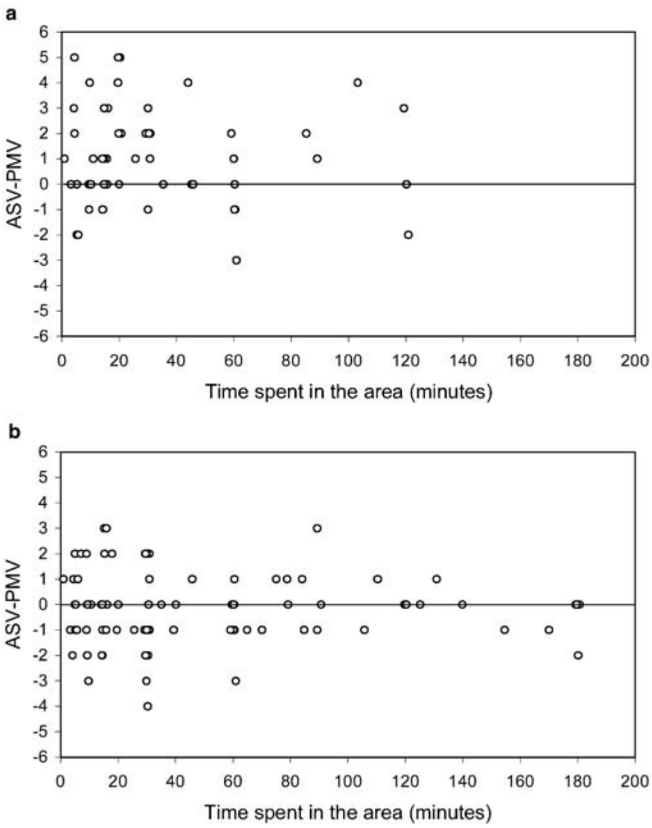


Fig. 12 Difference between thermal sensation of the interviewees (*ASV*) and thermal sensation assessed by the *PMV* index plotted against time spent in the area, when (*top*) ambient air temperature (T_a) $\leq$ 18.5 °C and (*bottom*) ambient air temperature (T_a) $\geq$ 18.5 °C. Positive values means that the subjects are warmer than actually predicted, negative values that subjects are colder than predicted and zero value that the thermal sensations of the interviewees are the same as predicted by the *PMV* index

Conclusions

It is found that the thermal environment, access and design are important factors in the use of the park. In order to continue to use the park when the thermal conditions become too cold or too hot for comfort, people improve their level of comfort consciously or unconsciously by modifying clothing and patterns of behaviour, choosing the most supportive thermal environment opportunities available within the place.

It is also found that people voluntarily expose themselves to warm conditions that are well outside theoretically acceptable comfort conditions when visiting the park. This implies that the park is a very special place in terms of thermal expectation and that psychological expectancy may influence subjective assessment and satisfaction. The fact that the park provides a variety of microclimates that offer a high degree of control and that people are there of their own free will further increase their tolerance and satisfaction.

In agreement with other field studies, the current results suggest that people undergo physical adaptation to improve the relation between the thermal environment and their requirements. They also suggest that psychological adaptation is important for the subjective assessment and enjoyment of the outdoor environment. Creating a variety of microclimates, i.e. sunny, shady, sheltered and exposed sub-spaces in outdoor public places, will increase both the physical and psychological adaptation. As a result, usage of outdoor spaces will be increased throughout the year.

The discrepancy between the thermal sensation experienced by the interviewees and the thermal sensation assessed by the PMV index indicates that steady-state models such as the PMV index may not be appropriate for the assessment of short-term outdoor thermal comfort. This is mainly due to their inability to represent transient exposure and psychological aspects, which are of particular importance to the subjective assessment of the outdoor environment.

This project is the first step in an attempt to achieve a better understanding of the implications of the thermal outdoor environment for human behaviour. Though the present study presents material from quite large-scale interviews, complementary studies in other urban environments, cities, countries and seasons will be conducted to further our understanding of the implications of the thermal environment on people's use of outdoor places.

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